Market Scale, Political Size, and Canada's Future

AI-enhanced research assignment — choose one project

Work in pairs • One week • Build an auditable argument • Canvas controls the exact deadline

Names _____ **Section** _____

The goal. Use economies of scale to answer a consequential Canadian policy question without treating a model as a verdict. Your pair will use an AI assistant as a research tool, verify its claims against primary sources, perform one transparent calculation, and make a recommendation whose assumptions and limits a reader can inspect. The submission deadline and upload location shown in **Canvas** are authoritative.

1. Choose one research question

	Project A: The Size of Nations	Project B: 100 Million Canadians
Decision	Compare three institutional arrangements for Canada and the United States.	Evaluate a scenario in which Canada's population reaches 100 million by 2100.
Core tension	Market scale versus the cost of making common policy when preferences differ.	Scale economies versus the housing, capital, infrastructure, and adjustment needed to realize them.
Required result	Identify the arrangement that best balances the two forces under your stated assumptions.	Identify where scale could plausibly raise productivity or variety, and where bottlenecks could prevent it.

Common economic mechanism. If producing a variety requires fixed cost F, average cost is $AC(q) = F/q + c$. A larger market may support a larger q, lower average cost, more viable varieties, or some combination of the three. This mechanism is a hypothesis to test, not proof that bigger is always better.

2. What every pair submits

1. A **600–800 word policy memo** with a clear question, mechanism, evidence, counterargument, and conclusion.
2. **One reproducible figure or table**: submit the data file and the code or exact spreadsheet steps that generated it.
3. **One simple calculation** with units, inputs, source, formula, result, and a sentence interpreting what it does—and does not—show.
4. A **five-claim evidence table** with direct URLs, access dates, and a short note explaining exactly what each source supports.
5. A **prompt log** containing the important prompts, substantive AI responses, corrections you made, and the model/tool used.
6. A short **division-of-work statement** signed by both partners.

AI is allowed; unverifiable AI prose is not evidence. You remain responsible for every number, quotation, citation, and inference in the memo. Never enter private student data, passwords, tokens, or other confidential information into a prompt.

Project A — The Size of Nations

An institutional-design counterfactual, not a proposal for political union

3. Analytical frame

Alesina and Spolaore ask how political borders balance benefits of size against the difficulty of choosing one policy for people with different preferences. Here, **political heterogeneity** means disagreement over public policy, taxes, regulation, redistribution, or public goods. It is not a ranking of cultures or identities. Economic integration may let sovereign countries share a large market without sharing all political decisions.

Your question. Under what conditions does each arrangement below best balance market access, policy autonomy, and the costs of political heterogeneity? Treat the comparison as a disciplined thought experiment. Do not argue that Canada or the United States *should* cease to exist.

Arrangement	Market access	Political authority	Costs or frictions to investigate
Separate protected markets	Mostly domestic	Separate	Tariffs, smaller customer base, duplicated fixed costs
Separate polities + CUSMA-style integration	Large shared market, subject to agreement rules	Largely separate	Rules of origin, border frictions, bargaining, policy spillovers
Hypothetical federation	Large internal market	More decisions pooled	Common-policy mismatch, representation, redistribution, constitutional transition

4. Required analysis

1. Explain the two-sided Alesina mechanism: why size can lower some per-person costs, and why preference differences can raise the cost of common policy.
2. Compare all three arrangements using the *same* criteria. At minimum: market scale, consumer variety, policy autonomy, trade frictions, distributional effects, and one factor the simple model omits.
3. Explain why an agreement such as CUSMA can partly separate **market size** from **political size**. Also explain why an integrated market is not literally frictionless.
4. Make a conditional recommendation: "Arrangement X performs best *if* assumptions A and B hold; otherwise Y may dominate."

Required calculation: fixed market-access cost per potential customer. Source current Canadian and U.S. populations. Let F be a common illustrative fixed cost. Compare

$$\frac{F}{N_{CAN}} \quad \text{with} \quad \frac{F}{N_{CAN} + N_{USA}}.$$

Report the ratio. Under the stylized zero-friction assumption, the integrated market and federation can have the same denominator but very different political arrangements. State why this is **not** a welfare estimate.

5. A useful first prompt

"Act as a skeptical research assistant. Build a comparison matrix for (1) protected separate Canadian and U.S. markets, (2) separate sovereign countries with CUSMA-style integration, and (3) a hypothetical federation. Use the Alesina–Spolaore size-of-nations framework. Separate market-scale benefits from political-heterogeneity costs. For each factual claim, give a primary-source URL and flag claims you cannot verify. Do not advocate political union."

Project B — 100 Million Canadians

A scale scenario, not a population forecast

6. Baseline and question

Statistics Canada estimated Canada's population at **41,798,407** on July 1, 2026. Reaching 100 million by 2100 is an advocacy scenario associated with the Century Initiative; it is **not** the Statistics Canada medium-growth projection and is not itself federal policy.

Your question. If Canada reached 100 million residents by 2100, through whatever lawful combination of demographic change and immigration you choose to state, where could a larger home market create scale benefits—and what investments or constraints would determine whether those benefits improve living standards?

7. Two calculations to understand; one must appear in the memo

Population path. Over 74 years, the constant compound growth rate is

$$g = \left(\frac{100{,}000{,}000}{41{,}798{,}407}\right)^{1/74} - 1 \approx 1.19\% \text{ per year.}$$

Compare this scenario with Statistics Canada's projection range. Do not describe either as a prediction.

Fixed cost per resident. If the same fixed cost F serves the whole population, then (with population measured in millions)

$$\frac{F/100}{F/41.8} = 0.418.$$

The arithmetic implies a 58.2% reduction in that particular fixed cost per resident. It does *not* imply a 58.2% fall in all costs: capacity, congestion, quality, geography, and new capital may change with population.

8. Required analysis

1. Choose **two scale-sensitive sectors** and explain the mechanism in each. Examples include specialized manufacturing, transport networks, research infrastructure, digital services, cultural industries, or advanced health services. Do not assume all sectors share the same technology.
2. Identify **two binding bottlenecks** that could block or reverse the gains. At least one must involve a stock that takes time to build, such as housing, transport, energy, water, schools, hospitals, or productive capital.
3. Distinguish throughout among **total GDP**, **GDP per capita**, and **welfare**. A larger population can raise total GDP mechanically; the important analytical question is whether productivity, real income per person, choice, and quality of life improve.
4. Explain one regional implication. A national total does not tell us where people, infrastructure, firms, or environmental pressures would be.
5. End with two measurable conditions that would make the scenario more likely to improve living standards.

9. A useful first prompt

"Act as a skeptical Canadian policy research assistant. Evaluate a scenario in which Canada's population reaches 100 million by 2100. Distinguish total GDP, GDP per capita, and welfare. Propose two sectors with plausible economies of scale and two bottlenecks that require investment. Compare the scenario with official Statistics Canada projections. Label the Century Initiative as an advocacy source. Give primary-source URLs, show calculations, and flag every claim you cannot verify."

AI Workflow and Evidence Standard

Use AI to accelerate inquiry, not to outsource judgment

10. A four-pass workflow

Pass	Purpose	What to ask AI to do	What you must do
1	Map	Define mechanisms, alternatives, and disputed claims.	Decide the question and comparison criteria.
2	Find	Suggest primary sources and direct URLs.	Open every link; reject broken, irrelevant, or secondary support.
3	Test	Recalculate values; generate counterarguments; find omitted variables.	Reproduce calculations and check units, dates, populations, and denominators.
4	Write	Critique structure and clarity after you draft.	Make every substantive judgment and preserve your own analytical voice.

11. Five-claim evidence table

Include this table in an appendix. Add rows if needed.

#	Exact claim in our memo	Primary source + direct URL	Access date	What the source actually establishes
1				
2				
3				
4				
5				

12. Verification rules

- **Traceability:** every empirical claim maps to a row in the evidence table or a precise citation in the memo.
- **Primacy:** use the statistical agency, treaty text/government assessment, or original research paper where available. An AI answer is never the source.
- **Date and definition:** verify the observation year, release date, units, geography, nominal/real status, and whether a value is a level, share, rate, scenario, or forecast.
- **Reproduction:** a classmate should be able to regenerate your figure/table from submitted files. Label sourced data separately from your calculations.
- **Uncertainty:** write "we could not verify" rather than filling a gap with a plausible-looking number. Separate evidence from inference.
- **Citation check:** open every URL after exporting the final PDF.

13. Prompts that improve rather than decorate the analysis

- "List the assumptions needed for my conclusion. Which is most fragile?"
- "Try to falsify my claim. What pattern in official data would count against it?"
- "Audit this table row by row. Recompute each statistic and flag mixed years, units, or denominators."
- "Is this a statement about total GDP, GDP per capita, productivity, or welfare? Rewrite it so the object is unambiguous."
- "Give the strongest counterargument in 100 words, then state what evidence would distinguish the two views."

Submission Map and Source Starter Pack

A compact package that another researcher can audit

14. Suggested memo architecture

Section	Guide	Job
Decision and answer	80–100 words	State the question, answer, and key condition.
Mechanism	150–180 words	Explain the scale mechanism and the central tradeoff.
Evidence and calculation	220–260 words	Interpret the figure/table and calculation; connect both to the mechanism.
Counterargument and limits	100–130 words	Give the strongest objection and what the model omits.
Conditional recommendation	70–100 words	Say what should follow under stated assumptions and what evidence could change your view.

The 600–800 word limit applies to the memo body, not the evidence table, references, prompt log, code, or division-of-work statement.

15. Assessment checklist

Before uploading, can both partners answer *yes* to each question?

- Is there a precise answer to the chosen research question rather than a list of possible effects?
- Does the memo correctly connect market size to fixed costs, output per variety, competition, or variety—without claiming scale solves everything?
- Does Topic A compare the same three arrangements neutrally, or does Topic B analyze two sectors, two bottlenecks, and GDP versus GDP per capita versus welfare?
- Are the calculation, units, figure/table, and direct-source claims reproducible and correctly interpreted?
- Does the counterargument engage the conclusion rather than merely add a generic caveat?
- Does the prompt log reveal how AI affected the work, including at least one correction or rejected suggestion?
- Does the PDF contain working links, and are all source and data/code files included in the Canvas upload?

16. Primary-source starter pack

Size of nations. Alesina and Spolaore, *The Size of Nations* (MIT Press): https://mitpress.mit.edu/9780262511872/the-size-of-nations/

Alesina, Spolaore, and Wacziarg, "Economic Integration and Political Disintegration," *American Economic Review* 90(5): https://www.aeaweb.org/articles?id=10.1257%2Faer.90.5.1276

CUSMA. Global Affairs Canada, economic impact assessment: https://international.canada.ca/en/global-affairs/corporate/reports/impact-assessments/2020-02-cusma-economic

Global Affairs Canada, statement following the 2026 joint review: https://www.canada.ca/en/global-affairs/news/2026/07/statement-by-minister-leblanc-following-trilateral-cusma-joint-review-meeting.html

Population. Statistics Canada, July 1, 2026 population estimate (released September 23, 2026): https://www150.statcan.gc.ca/n1/daily-quotidien/260923/dq260923b-eng.htm

Statistics Canada, population projections, 2025–2075: https://www150.statcan.gc.ca/n1/pub/17-20-0003/172000032026001-eng.pdf

Advocacy source—label it as such. Century Initiative: https://www.centuryinitiative.ca/

Access date for this starter pack: October 19, 2026. A starter source may orient your search, but it does not automatically support your particular claim. Cite the exact table, paragraph, or page you used. Add other primary sources as required by your argument.